

BCD

Binary-coded decimal stores each decimal digit in its own four-bit nibble

Four bits per digit

- Each BCD digit uses four bits and runs from zero through nine
- Nibbles A through F are invalid
- Packed BCD lays digits side by side in one word
- Do not confuse BCD with ordinary binary of the same decimal value

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

BCD encode / decode

Pack decimal digits into **4-bit nibbles**, spot invalid codes (A–F), and decode packed BCD back to decimal.

Starter example: Two-digit decimal 42 packs to BCD nibbles 4 and 2 → hex 0x42. Try an invalid nibble like 0x4A.

Load starter example

Challenges 0 / 22 cleared

Quiz: bits/digit: Bits per BCD digit? Answer: 4

Answer

Show hint

Check

Next

Idle

Quiz: bits/digit

Quiz: max digit

Quiz: invalid

Quiz: 42 packing

Starter 42

Encode 99

Encode 0

See invalid

Decode 42

Three-digit 255

Quiz: vs binary

Year 2026

Encode trace

Quiz: width bits

Round-trip

Need more digits

Quiz: MSD

Digit 9

Recover valid

Quiz: use

Hex path

Full BCD

Core ideas

One digit → 4 bits

BCD stores each decimal digit in a nibble (0–9 only).

Invalid codes

Nibbles A–F (10–15) are not decimal digits — decode fails.

BCD lab starter

Workbook practice

- In the workbook track, write packed BCD for forty-two, ninety-nine
- Mark why nibble A is illegal
- Encode two thousand twenty-six as four digits and check you get hex two-zero-two-six
- Note one pitfall: treating binary zero-x-two-A as if it were BCD for forty-two

Pitfalls to watch

- Do not mix binary and BCD encodings of the same number
- Invalid nibbles must be rejected, not silently accepted
- And remember: the browser lab is literacy
- Real displays and decimal datapaths still need correct packing width and digit checks

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, pack a few decimals and one invalid case by hand
- When you are ready, take the short quiz, then continue to parity and checksum